

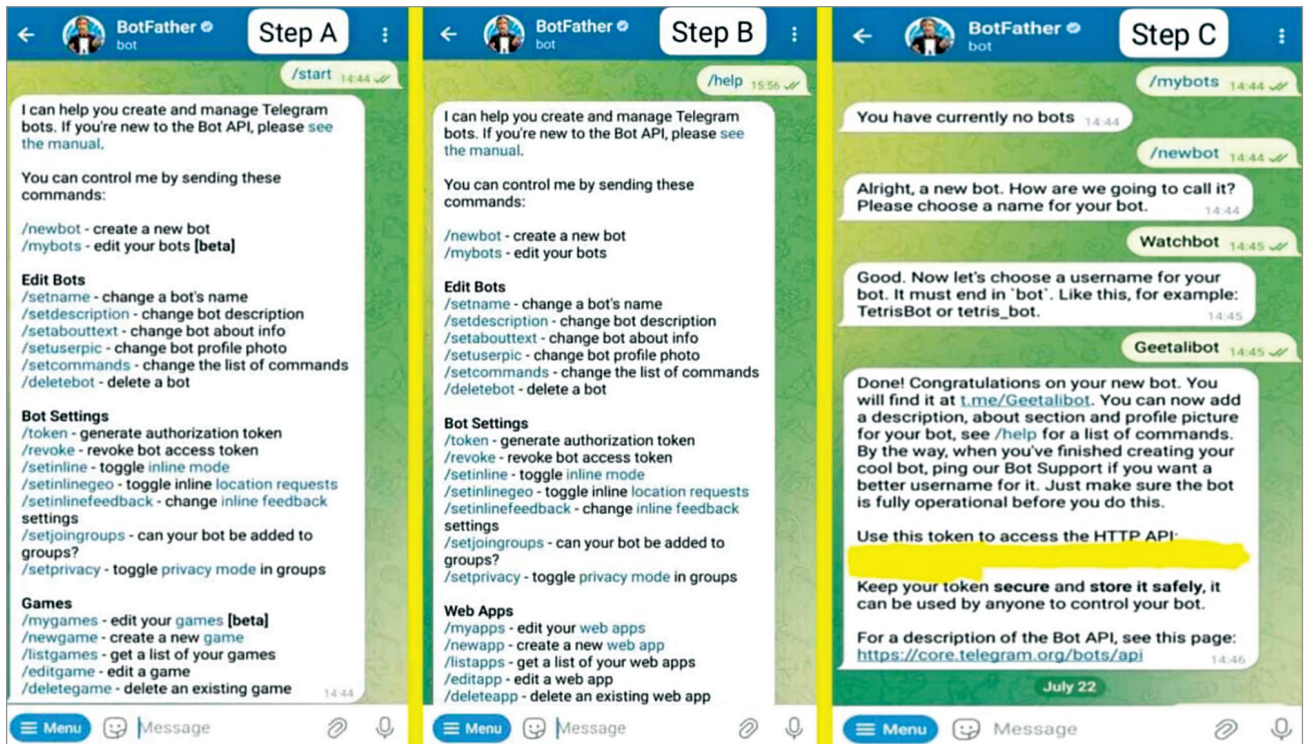


Fig. 6: Starting Telegram bot and getting the API

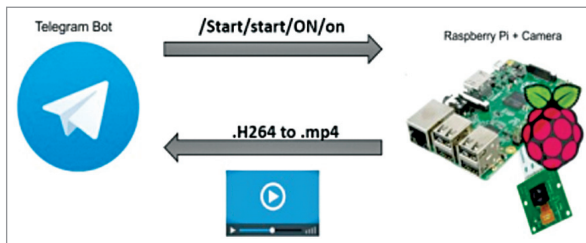


Fig. 7: Author's visualisation to obtain video through Telegram

```

import telepot
from picamera import PiCamera
import RPi.GPIO as GPIO
import time
from time import sleep
import datetime
from telepot.loop import MessageLoop
from subprocess import call

PIR = 7
camera = PiCamera()
camera.resolution = (640, 480)
camera.framerate = 25
GPIO.setwarnings(False)
GPIO.setmode(GPIO.BOARD)
GPIO.setup(PIR, GPIO.IN)
motion = 0
motionNew = 0
def handle(msg):
    global telegramText
    global chat_id
    chat_id = msg['chat']['id']
    telegramText = msg['text']
    print('Message received from ' + str(chat_id))
    if telegramText == '/start || /start || /ON || /on':
        bot.sendMessage(chat_id, 'Security camera is activated.')
    while True:
        main()
main()

```

Fig. 8: Code snippet importing library setting sensor pins and camera characteristics

our Telegram bot—Watchbot—can be activated using /start, /ON, or /on, as we've programmed it accordingly.

2. The Raspberry Pi, by default, records videos in .H264

format, which needs to be converted to .mp4, the most widely accepted video format.

To test the camera for capturing an image, use the following code snippet:

```

camera = PiCamera();
camera.start_preview()
camera.capture(path + '/pic.jpg',
              resize=(640,480))
time.sleep(2)
camera.stop_preview()
camera.close()
# Sending picture
bot.sendPhoto(id, open(path + '/pic.jpg',
                      'rb'))

```

At this stage, we can confirm that the camera is connected to the Raspberry Pi. Simultaneously, configure the Telegram settings. Fig. 7 provides a visualisation of obtaining video through Telegram.

Connecting (Raspberry Pi+ camera) to Telegram bot

Once the bot is created, the next step is developing the Python code